

Numpy
Pandas
Matplotlib

→ EDA - Titanic - usable

Numpy → foundation
↳ Data Science
AI Library

AI VL ML V8 DL

↳ Linear Regression
KNN, KMeans, DBSCAN

ANN - Intuition

Pandas - Numpy
Scikit - Numpy
Tensorflow, pytorch → Numpy

Numpy

100 Million Numbers

↳ List

numbers = [1, 2, 3, ...]

Integer → python object

List
- object
- object
- object

→ Value
dtype
Reference
memory

Numpy

↳ Directly

1, 2, 3, 4, 5
↳ Much faster
Use less memory
optimized 'C'

List → Heterogeneous

[1, Hello, True, 5.6]

Xlumpy array

One datatype

[1, 2, 3, 4, 5]

array = [1, 2, 3, 4]

Integer → float

Promoting
→ Safest Datatype

Type promotion

ndarray
→ Array-Numpy
type(array)

Dimensions → 0D, 1D, 2D, 3D, ND

'5'

a = np.array(5)

1D → (1, 2, 3, 4)

a = np.array([1, 2, 3, 4])

(4,) → Student Marks

2D Array

→ Rows x Columns

[90 80]
[70 60]

a = np.array([
[90, 80],
[70, 60]
])

3D Array

X, Y, Z

Page 1

1 2
3 4

Page 2

5 6
7 8

1000 Images

Xlumps

224 x 224 x 3
Height Width Channels (R, G, B)

1000 x 224 x 224 x 3

→ 4D Array

np.array([
[1, 2],
[3, 4]
],
[
[5, 6],
[7, 8]
])

Dimension

→ Physical Dimension

→ level of Indexing

Indexing Slicing

np.zeros(3,4)

↳ $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

image None placeholder

Identity matrix

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

np.eye(3)

↳ Linear, ML, Group

Reshaping

a = np.array([1,2,3,4,5,6])

↳ b = a.reshape(2,3)

1 2 3
4 5 6

↳ (copy data)
View

Random number

np.random.rand(3,2)

↳ Int - NN
Simulations

f.flatten

→

b.flatten

↳ 1D →

Copy

vs View

↳ Duplicate

↳ Show

Pandas → Dashboard

Python

[xlumpy]

[Pandas]

ML

Reading

Cleaning

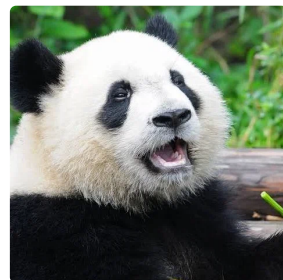
Fitting

Group

Analyze

Transforming

Exporting



df = Pd.DataFrame({
"Age": [10, 20, 30]
})

Pandas

↳ DataFrame

↳ Numpy array

↓
memory

faster

Company data

↳ csv - file

↳ Some →

60,000/

Average

Sort Age

Missing

Data Structure

(i) Series - [1D] →

(ii) DataFrame - [2D]

Table

Collection of Series

Name Age Salary
| | |
[Series]

Read → CSV, Excel, JSON, SQL

df = pd.read_csv(—)

Age, Year, City, Name

df['Age', 'Year', 'City']

a = df['Age']

↳ Type

↳ Series

b = df[['Age']] > 10

↳ Type

↳ DataFrame

8, 1, ~

age = pd.Series([20, 30, 40, 50])

print(age)

→

0 20
1 30
2 40
3 50
Index Values

ages = pd.Series([20, 50, 30], Index=[_, _, _])

5-Rows

↳ head()

df.head() → 5 Rows

↳ tail

df.tail()

Dataset information

df.info()

df.describe()

Selection

↳ i/loc → integer, index
loc → label

0 1 2
A B C

0 D 1 2 3
1 E 4 5 6

df.iloc[1, 1]

df.loc[E, B]

Add column

$df['Bonus'] = df['salary'] * 0.10$

50000 = 5000

Remove columns

→ $df.drop(columns=[''], inplace=True)$
(> new dataframe)

$df.isnull()$
- notnull

Pandas - Workflow

Read data

↓

Inspect
(head, info, describe)

↓

Clean
(dropna, fillna, rename)

↓

Transform
(m-e, filtering)

(> Analyse

(> groupby, agg, sorting

↓

VM - Visualization

Matplotlib

Computation

Data Manipulation

Visualization

Replace - 1000 - 10 Rows
(>

$df['Ase'] = df['Ase'].fillna(df$

$\Rightarrow df.isnull()$
 $= df.isnull().sum()$
 $= df.dropna()$

Sorting

$- df.sort_values('', asc$

Group
→ $df.groupby('') [sv -]$

- Line charts
- Bar
- Pie
- Histograms

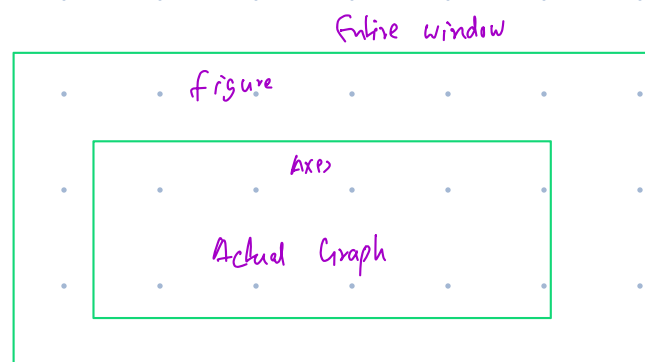
- Scatterplot
- Boxplot
- Heatmaps
- 3D Graphs

`import matplotlib.pyplot as plt`
 ↳ Behave - Canvas

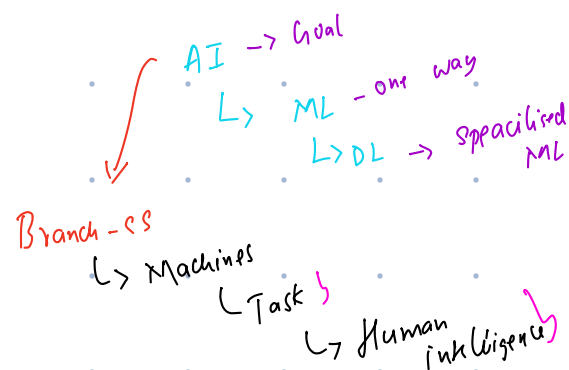
plt.show()
 ↳ Display - Canvas

figure → whole canvas
 Axes = plotting Area

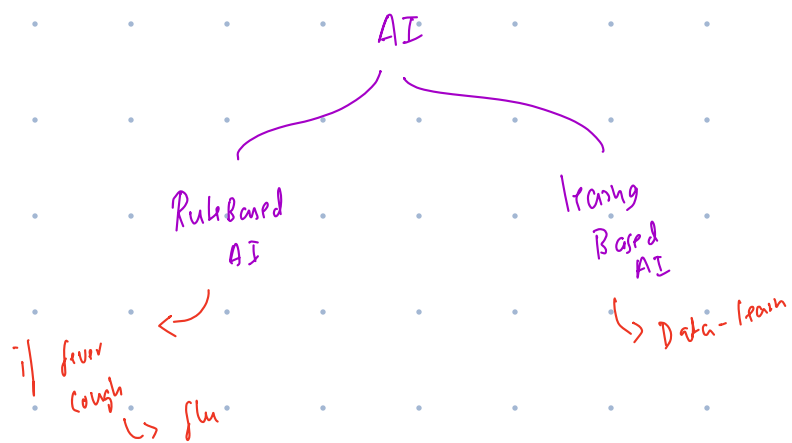
Figure Vs Axes



AI ML DL



understand
 Playing chess
 Driving
 Translation
 ChatGPT



Rule - Based

mimic - Dataset
(Get)

- **like / narrow**

- **General** -> Composable

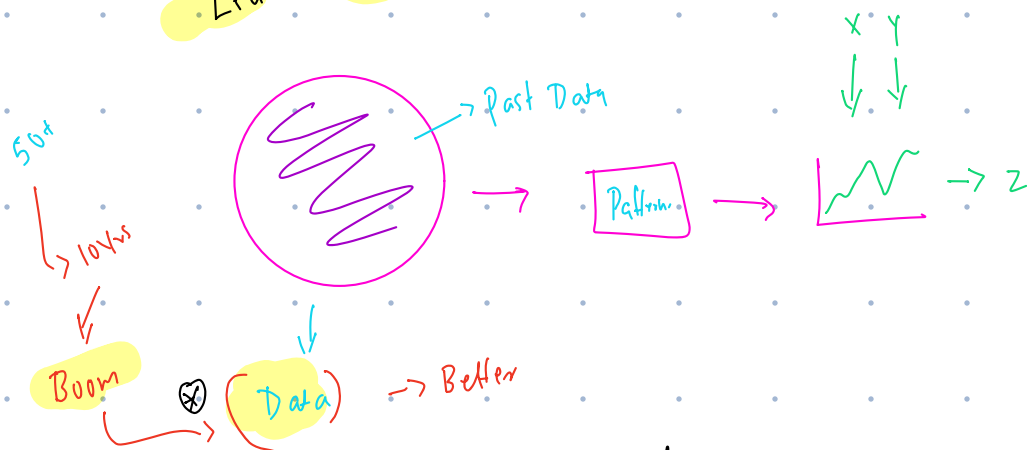
- **Strong AI** -> Beyond

ML -> Machine learning

(Data) - learn - forecast / probability



Learn - **without** explicitly programming



Data - 1 decade

* **Comput**

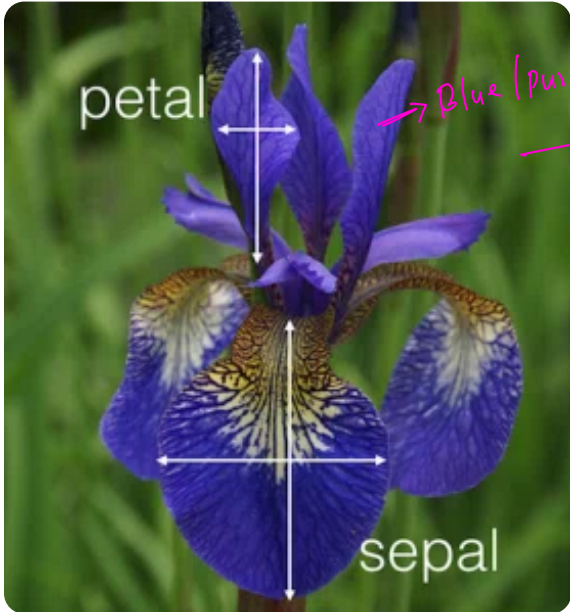
features

'x' Independent Variable

Target

- Dependent

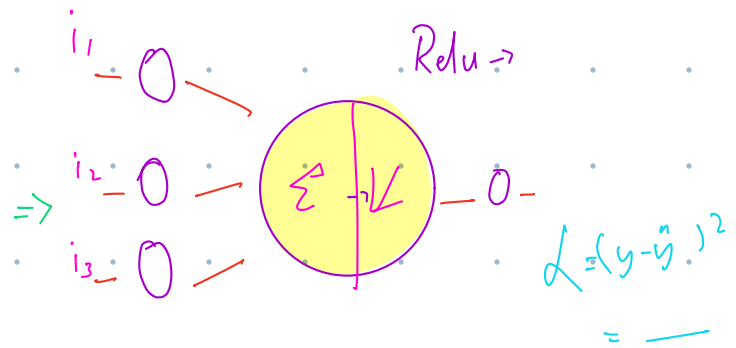
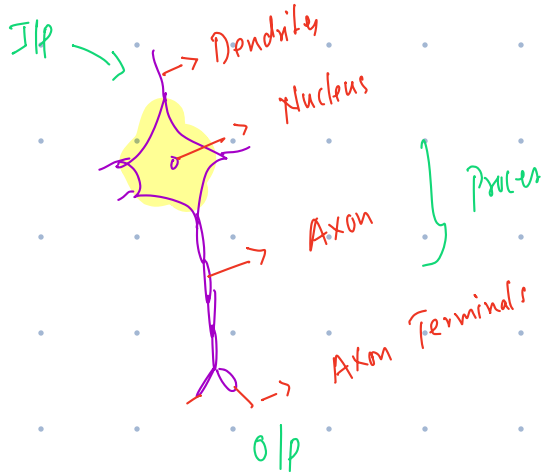
	sepal_length	sepal_width	petal_length	petal_width	species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa
7	4.6	3.4	1.4	0.3	setosa
8					



Types

- ✓ Supervised → Label + Data
 - ✓ Unsupervised → Unlabeled - Data
 - ✓ Reinforcement → feedback
- Feature Clusters

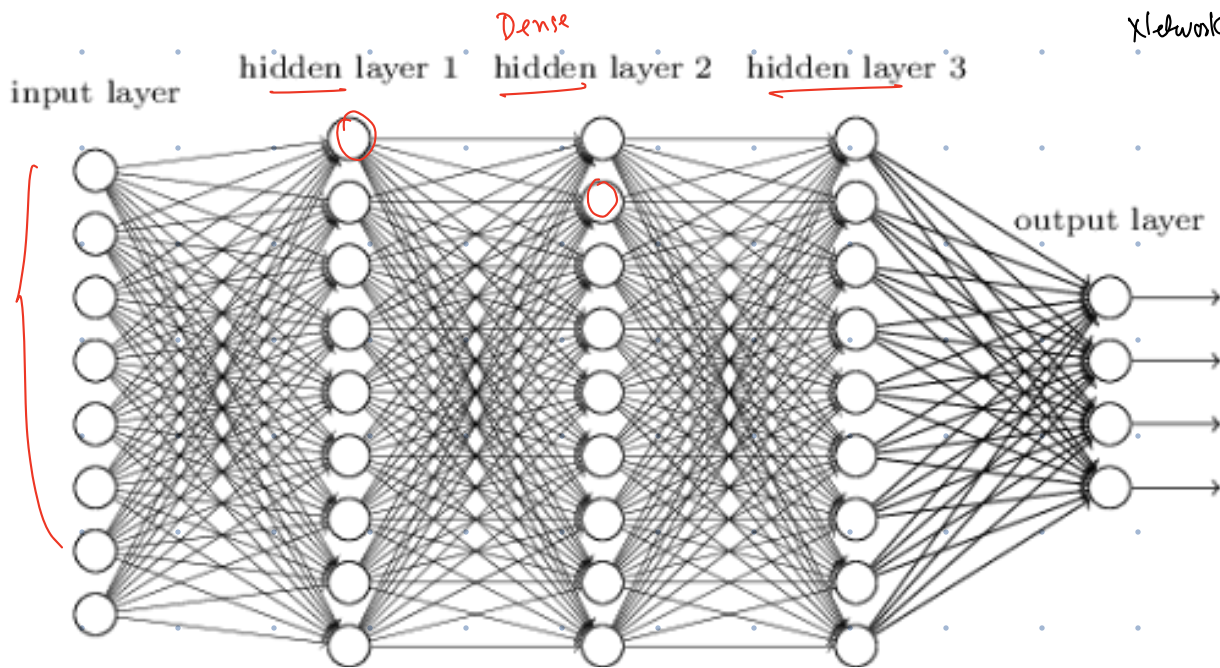
Deep learning



Perceptron

DNN

Deep Neural Network



Types

ANN - DNN — Dense
← CNN
RNN → Recurrent Neural Network — 1 layer
↳ LSTM
↳ GRU
↳ Attention

Deep Neural Network

GenAI
↳ LLMs